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EXECUTIVE SUMMARY

This report presents the Weather Impact Analysis System, a Python-based analytical prototype developed to examine how weather conditions vary across selected Nigerian locations and what operational and agricultural impacts can be observed within a simulated dataset. The system was built for a hypothetical organization, ClimateSmart Analytics Africa, which supports businesses, agricultural stakeholders, and community organizations affected by weather variability.

The analysis processes 30 simulated weather observations drawn from three locations: Lagos, Kano, and Ibadan, using core Python programming constructs including lists, dictionaries, conditional logic, loops, and custom functions. The system classifies rainfall and temperature levels, detects extreme weather events, calculates a custom weather-risk score, and compares locations based on their exposure to heat, rainfall, and operational disruption.

1.1 The Problem

Modern commercial agricultural operations face increasing operational and financial risks due to unpredictable climatic variations and severe weather events. In tropical agricultural zones across Nigeria, unmonitored fluctuations in ambient temperature, excessive rainfall and relative humidity directly threaten field activity schedules, crop yields and environmental stress thresholds, leading to unmitigated operational downtime, miscalculated field labor and unrecovered crop damage during peak disruption periods.

To address this challenge, commercial stakeholders require a structured data pipeline to ingest, validate and evaluate weather records. By transforming raw daily observations into quantifiable risk metrics, specifically identifying extreme heat days (temperatures $\geq 35^{\circ}\text{C}$), severe rainfall events and recorded disruption days, management can shift from reactive to proactive data driven operational planning.

1.2 The Analysis

To quantify the impact of weather fluctuations across key agricultural zones (Lagos, Kano, and Ibadan), a custom Python-based analytics pipeline was executed on 30 daily weather records spanning April and May 2025. The analytical framework made use of a custom validation and aggregation system without relying on python computational libraries.

The evaluation analyzed three primary dimensions of environmental impact across the regional datasets:

- **Environmental Data Aggregation:** Monthly baseline metrics were compiled for each location to track average temperatures, total rainfall, relative humidity levels, wind speeds, and crop yield outputs.
- **Extreme Event Identification:** Daily records were continuously evaluated against predefined physical threshold boundaries to identify critical environmental stress events. These included extreme heat days (temperatures $\geq 35^{\circ}\text{C}$), severe rainfall incidents (rainfall $\geq 100\text{mm}$), and recorded operational disruptions.

1.3 Major Findings

The data analysis across the 30 daily weather observations revealed significant regional climate variations and critical risk factors across the evaluated locations:

- **Regional Climate Variation:** The environmental risk profiles differed sharply by geography. Kano exhibited the highest vulnerability to temperature stress, recording the most frequent extreme heat events (temperature $\geq 35^{\circ}\text{C}$). Conversely, Lagos experienced the highest cumulative rainfall and frequency of heavy precipitation incidents ($\geq 100\text{mm}$). Ibadan demonstrated a balanced micro-climate with moderate temperature and rainfall ranges, yielding the most stable crop outputs.
- **Periodic Risk Analysis (April vs. May):** A comparative evaluation showed an increase in weather extremes transitioning from April to May 2025. May recorded a higher frequency of extreme weather events resulting in a marked increase in total operational disruption days compared to April. Further investigation is required to determine whether the weather extremes are a causal factor.
- **Location Risk Analysis (Kano vs Lagos vs Ibadan):** Kano recorded the greatest heat exposure, while Lagos and Ibadan recorded greater rainfall-related disruption. All ten records from Kano were instances of extreme events. Fifteen operational disruptions were recorded across the sample frequently coinciding with heavy rainfall or elevated risk scores.
- **Operational Disruption and Yield Impact:** Operational disruptions ("Yes" status days) were frequent around dates with extreme weather, specifically high rainfall coinciding with high humidity or extreme thermal spikes. These disruption days directly impacted field productivity, leading to lower average crop yields during highly disrupted periods.

1.4 Recommendations

Based on the empirical findings from the weather observation analysis, farm management should implement the following targeted strategic interventions to mitigate climatic risks:

- **Regional Infrastructure Alignment:** Kano operations require immediate investments in heat-tolerant crop varieties and shade-netting infrastructure to buffer against extreme thermal spikes. In Lagos, management must prioritize high-capacity drainage channels and raised seedbed systems to minimize field flooding during peak rainfall days.
- **Dynamic Labor Scheduling:** High-labor field activities such as fertilizer application, pesticide spraying, and harvesting must be dynamically scheduled ahead of predicted heavy rainfall and high-humidity events to eliminate input wastage.

BUSINESS PROBLEM

2.1 Background

Weather conditions influence a wide range of economic and operational activities, including agricultural planning, business continuity, transportation, and disaster preparedness. In Africa, where many sectors remain highly sensitive to weather variability, the ability to convert raw weather observations into actionable insight carries significant value for decision-makers.

Organizations such as ClimateSmart Analytics Africa are tasked with helping stakeholders understand how weather conditions vary across locations and what potential operational and agricultural consequences may follow. However, raw weather records consisting of temperature, rainfall, humidity, wind speed, crop yield, and disruption data are difficult for non-technical decision-makers to interpret in their unprocessed form.

This project addresses that gap by developing a lightweight analytical system capable of summarizing weather patterns, classifying conditions, flagging extreme events, and generating a simple risk indicator. While the system does not replace official meteorological forecasting bodies such as the Nigerian Meteorological Agency (NiMet), it demonstrates how a data professional can apply foundational programming skills to support better-informed decisions in weather-sensitive sectors.

2.2 Operational Challenges

The primary business problem stems from three critical operational vulnerabilities:

- **Unmonitored Thermal and Hydrological Stress:** Crop physiology reacts to extreme environmental thresholds. Temperatures exceeding 35°C accelerate crop transpiration and heat stress, while heavy rainfall (reaching or exceeding 50mm – 100mm) saturates field soils. Without automated threshold detection, management cannot intervene early enough to protect crop health.
- **Preventable Field Downtime:** Operational disruptions such as machinery breakdown, flooded transport routes, or hazardous working conditions often occur unexpectedly. Failing to track disruption patterns against specific weather indicators prevents field supervisors from proactively adjusting labor schedules or protective measures.
- **Financial misalignment:** Farm management currently lacks a direct mechanism to quantify how localized weather volatility impacts revenue. Without linking weather indicators, disruption frequency, and harvested yield volumes into a single analytical framework, budgeting and financial risk forecasting remain largely speculative.

To address these vulnerabilities, this project establishes a Python data retrieval, validation, and analytics system designed to achieve the following specific business objectives:

1. **Retrieve Data:** Process raw weather observations (covering April and May 2025 across Lagos, Kano, and Ibadan).
2. **Quantify Regional and Monthly Risk:** Evaluate average environmental conditions, count extreme weather occurrences, and measure total disruption days across geographic locations and monthly observation windows.
3. **Provide Actionable Business Visibility:** Deliver structured reports to empower farm operations managers to query extreme events, assess crop yield impacts, and optimize field resource allocation.

DATASET

3.1 Data Size

The dataset comprises 30 weather observation records collected across the target agricultural locations for April and May 2025. Each observation record is represented as an individual dictionary inside a list for iterative Python processing without third-party database dependencies.

3.2 Locations

Data collection targets three distinct Nigerian locations: Lagos, Kano, and Ibadan (10 observations per location) to capture geographical micro-climate diversity:

- **Lagos:** Characterized by elevated humidity, high cumulative precipitation, and frequent operational disruptions.
- **Kano:** Defined by high ambient temperatures, frequent thermal stress events, and lower cumulative seasonal rainfall.
- **Ibadan:** Exhibits moderate precipitation and temperature profiles, serving as a baseline for stable agricultural productivity.

3.3 Variables

Each observation record within the dataset contains eight parameters tracking physical environmental conditions, operational status, and crop productivity:

- Date (YYYY-MM-DD): str, observation date
- Location: str, city where the observation was recorded
- Temperature: float/ int, recorded temperature in degrees Celsius
- Rainfall: float/int, recorded rainfall in millimeters
- Humidity: float/int, relative humidity as a percentage
- Wind Speed: float/int, recorded wind speed in kilometers per hour
- Crop Yield: float/int, field output in tons per hectares
- Disruption: whether operational disruption occurred ("Yes"/"No")

3.4 Data Source

The data does not originate from official meteorological sources and should not be presented as representing actual historical weather conditions. It was designed solely to reflect the structure of a real-world weather-impact analysis problem for the purpose of practicing Python-based analytical thinking, in line with the study group's progressive learning approach of mastering core logic before introducing external data-import tools such as Pandas or CSV/API integrations.

METHODOLOGY

The system is built using foundational Python programming logic without relying on external data-import or data-manipulation libraries like Pandas, NumPy, or Matplotlib. The core architectural methods include:

4.1 Data Presentation

The analysis utilizes a simulated educational dataset structured as a Python list containing dictionaries, where each dictionary represents an individual weather observation row analogous to an Excel table format.

4.2 Record Processing

Data records are processed iteratively using Python for loops, conditional statements, comparison operators, and arithmetic operations to calculate counts and aggregates. The system operates through an imperative sequence:

1. **Retrieval:** Raw records are iterated over sequentially using standard Python for loops.
2. **Type Enforcement:** Numeric string fields from user inputs are converted to float or integer primitive types.
3. **Boundary Verification:** Each numeric value is validated against predefined physical range limits (e.g., non-negative checks for rainfall and wind speed; temperature sanity checks). Out-of-bounds records are flagged before aggregation.
4. **Accumulation:** Valid records feed directly into dedicated functions and loops to track regional and monthly totals.

4.3 Weather Classification

Custom Python functions (`classify_rainfall()` and `classify_temperature()`) evaluate observations based on thermal and hydrological thresholds:

- Thermal Thresholds:
 - Cool Temperature: Below 25 °C
 - Moderate Temperature: 25 – 30 °C
 - Hot: 31 - 35 °C
 - Extreme Heat: Above 35 °C
- Hydrological Thresholds
 - No Rain: Rainfall = 0 mm

- Light Rainfall: 1 -10 mm
- Moderate Rainfall: 11 – 50 mm
- Heavy Rainfall: 51 – 100 mm
- Extreme Rainfall: Above 100 mm

4.4 Extreme Events Identification

The system identifies and isolates extreme weather events using Boolean conditional logic during loop execution. A record is flagged as an extreme event if it meets or exceeds target operational parameters:

- **Extreme Thermal Event:** Triggered when temperature ≥ 35 °C
- **Extreme Rainfall Event:** Triggered when rainfall ≥ 100 mm.
- **Compound Extreme Event:** Triggered when an extreme thermal or precipitation flag coincides with a recorded operational disruption.

4.5 Risk Calculation

A custom risk-scoring model (calculate_risk_score() and classify_risk()) assigns points based on rainfall, temperature, and operational disruption criteria, classifying the final score into Low, Moderate, High, or Critical risk levels.

FINDINGS

5.1 Regional Climatic Analysis

Execution of the analytical scripts across the 30-observation dataset yields the following primary insights:

- **Highest Recorded Temperature:** The maximum temperature recorded across the dataset is 41°C, observed in Kano on 2025-05-10.
- **Highest Recorded Rainfall:** The maximum rainfall recorded is 110 mm, observed in Lagos on 2025-05-20.
- **Location Temperature and Rainfall Extremes:** Kano exhibits the highest heat exposure and temperatures, while Lagos and parts of Ibadan experience higher heavy rainfall and operational disruption exposure.
- **Disruptions and Extreme Events:** A total of 15 operational disruptions were recorded across the sample, frequently aligning with heavy rainfall events or high-risk scores.
- **Crop Yield and Weather Patterns:** Within this simulated sample, higher temperatures and extreme weather conditions are observed alongside variations in crop yield and operational disruption.

5.2 Comparative Monthly Analysis (April vs. May 2025)

A temporal comparison between April and May demonstrates a shift in climatic volatility transitioning into the main rainy season in southern Nigeria and peak heat in the north:

- **Thermal Trajectory:** Average temperatures dropped slightly from April (33.1°C) to May (31.8°C). However, extreme heat days in Kano remained clustered in late April and early May.
- **Hydrological Acceleration:** Total rainfall increased significantly from April (160 mm across all sites) to May (280 mm across all sites), driven by intensifying storms in Lagos and Ibadan.
- **Operational Disruption Shift:** Total disruption days rose from 5 days in April to 9 days in May, directly reflecting the impact of heavy precipitation on transportation, field accessibility, and scheduled farm operations.

RECOMMENDATIONS

Based on the findings derived from the analysis, decision-makers and organizational stakeholders should consider the following practical measures:

- **Heat-Related Contingency Planning:** Organizations operating in heat-sensitive environments like Kano should incorporate robust heat mitigation and contingency strategies such as artificial irrigation, shade cropping to reduce canopy solar radiation, and organic mulching to regulate soil moisture to buffer against high thermal exposure.
- **Genetic Vulnerability Mitigation:** Address the observed correlation between high temperatures and reduced yields by transitioning to heat-tolerant crop varieties engineered to withstand thermal stress during critical growth phases.
- **Rainfall Disruption Protocols:** Stakeholders in southern locations prone to heavy rainfall should establish early-action protocols for weather-induced operational disruptions.
- **Targeted Location Infrastructure:** Construct high-capacity drainage systems in heavy rainfall areas (Lagos and Ibadan) to prevent waterlogging and field access delays, while building water retention and storage infrastructure in high-temperature areas (Kano) to ensure water security.
- **System Scalability:** Management should consider expanding the current Python prototype to ingest larger datasets and real-world meteorological data feeds as analytical needs grow.
- Based on these findings, the report recommends heat-mitigation planning for heat-sensitive locations, early-action protocols for rainfall-prone areas, and future expansion of the system to incorporate real-world meteorological data.

LIMITATIONS

While the weather impact analysis prototype successfully meets project objectives, several inherent constraints should be acknowledged:

- The analysis relies on a fixed, small-scale simulated dataset of only 30 observations across three locations, which limits broader geographical generalizability and statistical depth.
- By strictly adhering to core Python structures (lists and dictionaries) without leveraging advanced data analytics libraries like Pandas or NumPy, complex multi-variable modeling and deep statistical aggregations are constrained.
- The system processes static historical entries and does not currently support real-time meteorological API integrations or live streaming weather feeds.
- The dataset is restricted to a two-month observation window (April and May 2025). This short-term timeframe captures only seasonal micro-trends and excludes any long-term evaluation of annual weather cycles critical for crop production.
- The data lacks site-specific precision. It aggregates information across broad state-level boundaries (e.g., Kano and Lagos). This spatial over-generalization fails to account for local microclimates and soil variation across individual farms.
- Important variables such as operational disruption were captured as Boolean classifications (Yes/No). This structure omits critical quantitative metrics, including the exact duration of downtime and the associated financial or yield losses.
- Critical soil and crop health indicators, including soil moisture levels and specific crop growth stages, were absent from the dataset, limiting agronomic evaluation.

FUTURE IMPROVEMENT

To build upon the current prototype and scale its capabilities for professional deployment, future development phases should incorporate the following enhancements:

- Migrate from hardcoded Python dictionaries to relational database systems (such as SQLite or PostgreSQL) or CSV file ingestion to handle thousands of historical records seamlessly.
- Expand data collection across broader geographic regions while capturing local, farm-level coordinates. Integrating site-specific locations alongside comprehensive soil and agronomic parameters will allow the system to deliver actionable, localized insights directly to farmers.
- Transition of the primary data storage format from nested dictionaries to structured tabular formats (e.g., DataFrames, CSV, or Excel). This restructuring will streamline and improve efficiency and downstream processing as record volumes scale.
- Incorporate validated, real-time meteorological feeds from official agencies such as the Nigerian Meteorological Agency (NiMet) and other accredited weather services.
- Integrate data science libraries like Pandas for data manipulation, and Matplotlib or Seaborn to generate automated visual trend charts directly from the analysis.
- Expand future data collection to track parameters like soil moisture and aggregate environmental metrics by specific crop growth stages, enabling precise resource planning tailored to the crop's physiological needs at each phase.

Connect the script to live weather forecasting APIs to pull real-time regional meteorological metrics, enabling automated alerts and predictive risk modeling.